

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Dynamic equilibria, feedback & tipping points - Set 1

Mark scheme

AQA spec 3.6.1; Sustainability; textbook pp. 370-372

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark.</i> • Development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs (1)	1
01.2	<i>1 mark for each point.</i> • Impact depends on total population x per capita impact (1) • With few people, natural processes (forest regrowth, fish recruitment, aerobic decomposition / dilution) recover faster than the resource is used or the waste added (1) • Larger populations exceed the rate of natural recovery: forest and soil do not regenerate, catches exceed MSY, water deoxygenates (1)	3
01.3	<i>1 mark for each point.</i> • Urban areas are supported by resources brought in from elsewhere; when one area is depleted, exploitation simply moves (timber, fish, minerals, energy) so consumers see no shortage (1) • Rural communities exploit local resources directly so the effects of over-exploitation become obvious quickly, encouraging cultural limits to need (1)	2

Question	Marking guidance	Marks
01.4	<i>Up to 4 marks in total; award marks in proportion to the completeness of each point.</i> • Climate: warming increases evaporation, cloud and albedo; higher CO₂ increases photosynthesis and sequestration; hydrological cycle: more evaporation gives more precipitation; populations: density-dependent factors return numbers to carrying capacity (up to 2) • In agriculture soil degradation, water shortage and loss of predators would reduce yields (negative feedback limiting the activity) (up to 2) • Humans respond with fertilisers, irrigation and pesticides to maintain yields, further damaging the natural systems that support production (up to 2)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-372; AQA spec 3.6.1.	
02.1	<i>1 mark for each point.</i> • Negative feedback responds to any change by re-establishing the previous equilibrium (1) • Only large unusual events overwhelm the feedback so a new equilibrium forms (1) • Example: methane hydrate release at the end of the Permian causing climate change and mass extinction (1)	3
02.2	<i>1 mark for each point.</i> • In a diverse system each component is relatively less important so the loss / change of one has a small effect; alternative food sources and relationships buffer the change (1) • Natural: coral reefs and tropical rainforests with complex food webs are stable; simple polar / desert systems fluctuate (1) • Human: agroecosystems focus on a few crops and remove detritivores, pollinators and predators so are vulnerable to pests, disease, weather (1) • Diverse technical systems: a society using a range of renewable energy resources is more secure than one relying on a single source (1)	4
02.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Melting permafrost releasing methane (1) • Reduced albedo as ice / snow melt (1) • Methane hydrate release (1) • Forest and peat fires (1) • Faster soil decomposition (1) • Ocean acidification reducing sequestration (1) • Cirrus cloud formation (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 371-372; AQA spec 3.6.1.	
03.1	<i>1 mark for each point.</i> • Moral belief: protecting the environment is the 'right' thing to do regardless of benefit to people (1) • Enlightened self-interest: protecting the environment is done because it causes fewer difficulties (for people) than not doing so (1)	2
03.2	<i>1 mark for each point.</i> • Total impact = population size x the individual (per capita) impact of each person (1) • Per capita impact rises with affluence / access to energy, so both a larger population and a more affluent population increase the total impact (1)	2

Question	Marking guidance	Marks
03.3	Any 1 from the points below - 1 mark each, maximum 1. • Slash and burn agriculture (1) • Fishing (1) • Discharge of sewage / biodegradable pollutants (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370; AQA spec 3.6.1.	
04.1	1 mark for each point. • Negative (1) • Positive (1) • Negative (1) • Positive (1)	4
04.2	1 mark for each point. • The response to a change acts to re-establish / return the system to the previous equilibrium (1) • So the change does not become self-sustaining unless a large unusual event overwhelms the negative feedback (1)	2
04.3	1 mark for each point. • eg the possible release of methane hydrate at the end of the Permian era (1) • Overwhelmed the negative feedback mechanisms, causing climate change and a mass extinction / a new equilibrium (1)	2
04.4	1 mark for each point. • Soil degradation, reduced water availability and loss of natural pest predators would reduce yields (negative feedback curtailing the activity) (1) • Instead of reducing the activity, farmers use fertilisers, irrigation and pesticides, which further damage the natural systems that support food production (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 371-372; AQA spec 3.6.1.	
05.1	1 mark for each point. • The new opportunity is overwhelmingly attractive (economic benefit) (1) • Lack of knowledge / understanding of the impacts (1) • The impacts occur elsewhere or much later, so are not evident to those who benefit (1)	3
05.2	1 mark for each point. • Cultural belief that resources should be exploited to satisfy needs only (1) • Problems of over-exploitation become obvious quickly as local supplies dwindle, so behaviour changes (1)	2
05.3	1 mark for each point. • Resources are brought in from other areas (1) • If one area is depleted, exploitation moves to another area (eg timber, fish, metals, energy) without the consumers noticing (1)	2
05.4	1 mark for each point. • Easier access to energy has made many new activities possible (1) • The scale of activities has greatly increased (1) • Toxic wastes are produced that cannot be decontaminated by natural processes (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-371; AQA spec 3.6.1.	

Question	Marking guidance	Marks
06.1	1 mark. Development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs (1)	1
06.2	1 mark for each point. - Negative feedback resists change / returns the system towards its previous state (1) - eg increased temperature increases cloud cover and albedo, or higher CO₂ increases photosynthesis (1) - Positive feedback increases the change / makes it self-sustaining (1) - eg melting permafrost releasing methane, declining albedo as ice melts, forest and peat fires (1)	4
06.3	1 mark. Many species with a complex food web so the loss of one has a small effect; agroecosystems have few species and lack detritivores, pollinators and pest predators that maintain stability (1)	1
06.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) - Principles of natural systems: dynamic equilibria and negative feedback; diverse systems resistant to change; low energy-density renewable energy (solar-driven hydrological, carbon and nitrogen cycles); low-temperature processes using enzymes; material cycles with wastes as raw materials; small number of abundant elements; biodegradable non-toxic wastes - Human systems: non-renewable energy, high temperatures (Haber process, incineration, manufacturing), linear resource use, dispersal of minerals and mixtures that are hard to separate, persistent toxins (PCBs, organochlorines, toxic metals in electronics) - Applications: renewable energy and heat pumps, bioleaching, recycling and design for re-use, industrial symbiosis, organic and integrated agriculture, retaining hedgerows and pollinators, circular economy - Human responses that override feedback (fertilisers, irrigation, pesticides) and positive feedback triggered by human activity (permafrost, methane hydrates, fires); tipping points - Barriers: population size and affluence, attitudes of urban populations to distant resources, economics, need for international cooperation, existing infrastructure - Justified conclusion on how far natural principles can be applied	9
	Model answer - Level 3 (8 marks)	

Question	Marking guidance	Marks
	Natural systems have remained sustainable because they run on low energy-density renewable energy, mainly sunlight driving the hydrological, carbon and nitrogen cycles; their reactions occur at low temperatures because enzymes lower activation energies; materials cycle so the waste of one process is the raw material of the next; only a few abundant elements are used; and wastes are biodegradable and non-toxic. Negative feedback keeps them in dynamic equilibrium and high diversity makes them resistant to change.	
	Human systems contrast on every point: non-renewable fossil fuels, high-temperature processes such as the Haber process and incineration, linear use that disperses minerals and mixes materials that cannot be separated, and persistent toxins like PCBs and the metals in electronics. We also override negative feedback - when degraded soil reduces yields we add fertiliser, irrigation and pesticides - and trigger positive feedback such as permafrost melting.	
	Copying natural principles is already possible in many areas: renewable energy and heat pumps, bioleaching at low temperature, recycling and design for disassembly, industrial symbiosis, legumes and organic matter instead of manufactured nitrate, and retaining hedgerows for pest predators - the basis of the circular economy. The limits are population size and affluence, the attitude of urban consumers who never see depleted resources, cost, and the need for international cooperation. So the principles can make individual activities far more sustainable, but only a large-scale, cooperative shift in lifestyle would make the whole human system sustainable.	
	Weaker answer - Level 2 (5 marks)	
	Natural systems use renewable energy from the Sun and recycle materials, so nothing runs out and there is no waste. Humans use fossil fuels which will run out and produce waste that is toxic. Natural processes happen at low temperatures but the Haber process needs high temperatures. Humans could copy natural systems by using renewable energy like solar and wind, recycling more and using less fertiliser. This would make human activities more sustainable. However it would be expensive and people would have to change their lifestyle.	
	What makes the difference	
	• Level 2 states the contrast correctly but does not explain the mechanisms (enzymes, negative feedback, diversity, dispersal of minerals) that Level 3 uses.	
	• Level 3 applies the principles to specific technologies (bioleaching, heat pumps, industrial symbiosis, legumes); Level 2 stays at 'recycle more'.	
	• 'Evaluate the extent' - Level 3 distinguishes individual activities from the whole system in its conclusion; Level 2's evaluation is a single sentence about cost.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-374; AQA spec 3.6.1-3.6.3.	